

Supplemental Material

A Time-Aware Prompting Details and Analysis

During the inference of unaligned and normally-aligned LMs, we add five time-insensitive few-shot examples in the prompt as shown in the third row of Table 7 (denoted as “insensitive 5-shot without time”) to ensure that the LMs would respond to questions with short answers. For time-aware prompting, we propose that we can either specify the year information explicitly in the prompt (Table 8), or use a few time-sensitive examples in the prompt with answers from the target year (Table 9). Our final strategy uses both types of information (Table 10).

We analyzed the performance of different time-aware prompting strategies applied to LLaMa2-70B models with the target years 2021 and 2022 in Table 6. In general, **all prompting techniques improve the model’s performance** in both target years, indicating that either time-sensitive examples or time (year) mention help LMs elicit temporal knowledge better. In addition, using time-sensitive examples plus the year-mention in the prompt achieves higher target-year and temporal-decayed performance. Meanwhile, **prompting LMs to be aligned to 2022 causes more performance degradation on $\mathcal{F}_{\max}$** , suggesting that prompting LMs to be aligned to a more recent year could lead to more QA errors.

Prompting Strategy	$\mathcal{F}^{2022}$	$\mathcal{F}_{\text{decay}}^{2022}$	$\mathcal{F}^{2021}$	$\mathcal{F}_{\text{decay}}^{2021}$	$\mathcal{F}_{\max}$
Insensitive 5-shot w/o time	17.2	34.1	19.8	37.0	55.9
Insensitive 5-shot w/ time to 2022	24.4 (+7.2)	37.9 (+3.8)	24.4 (+4.6)	39.1 (+2.1)	52.8 (-3.1)
Sensitive 5-shot w/o time to 2022	27.6 (+10.4)	40.2 (+6.1)	22.8 (+3.0)	39.9 (+2.9)	53.6 (-2.3)
Sensitive 5-shot w/ time to 2022	27.4 (+10.2)	40.3 (+6.2)	23.9 (+4.1)	40.5 (+3.5)	54.1 (-1.8)
Insensitive 5-shot w/ time to 2021	19.2 (+2.0)	36.8 (+2.7)	25.3 (+5.5)	39.9 (+2.9)	53.7 (-2.2)
Sensitive 5-shot w/o time to 2021	19.4 (+2.2)	38.5 (+4.4)	25.0 (+5.2)	41.9 (+4.9)	56.8 (+0.9)
Sensitive 5-shot w/ time to 2021	19.7 (+2.5)	38.9 (+4.8)	25.9 (+6.1)	42.4 (+5.4)	56.7 (+0.8)

Table 6: Performance of different time-aware prompting strategies applied to LLaMa2-70B models on the TAQA test set. Same as §4.2, we evaluate the LMs based on the target-year ($\mathcal{F}^{2022}$ and $\mathcal{F}^{2021}$), temporal-decayed ($\mathcal{F}_{\text{decay}}^{2022}$ and $\mathcal{F}_{\text{decay}}^{2021}$), and historical-max ($\mathcal{F}_{\max}$) F1 scores, as defined in §2.3.

Prompt: Answer the following question: What is the capital of France? The answer is: Paris Answer the following question: Who wrote Harry Potter? The answer is: J.K. Rowling Answer the following question: Where did the Titanic sink? The answer is: Atlantic Ocean Answer the following question: What is the gravity of earth? The answer is: 9.807 m/s^2 Answer the following question: Is the speed of light faster than the speed of sound? The answer is: Yes Answer the following question: Which team won the UEFA Europa League? The answer is: Expected Output: Eintracht Frankfurt

Table 7: Our prompt for prompting the pretrained LM using the insensitive w/o time strategy for time-sensitive QA tasks.

Prompt: Answer the following question: What is the capital of France? As of year 2022, the answer is: Paris Answer the following question: Who wrote Harry Potter? As of year 2022, the answer is: J.K. Rowling Answer the following question: Where did the Titanic sink? As of year 2022, the answer is: Atlantic Ocean Answer the following question: What is the gravity of earth? As of year 2022, the answer is: 9.807 m/s^2 Answer the following question: Is the speed of light faster than the speed of sound? As of year 2022, the answer is: Yes Answer the following question: Which team won the UEFA Europa League? As of year 2022, the answer is: Expected Output: Eintracht Frankfurt
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Table 8: Our prompt for prompting the pretrained LM using the insensitive w/ time strategy for time-sensitive QA tasks.

Prompt: Answer the following question: Which Hindi film has the highest domestic net collection currently? The answer is: Brahmāstra: Part One – Shiva Answer the following question: Where is the NHL Winter Classic taking place? The answer is: Target Field Answer the following question: Who are the current drivers for the Mercedes-Benz Formula One team? The answer is: Lewis Hamilton George Russell Answer the following question: Who received the Player of the Game award for offense in the most recent Rose Bowl Game? The answer is: Jaxon Smith-Njigba Answer the following question: Where was the final of the last FIFA Club World Cup held? The answer is: Prince Moulay Abdellah Stadium, Rabat Answer the following question: Which team won the UEFA Europa League? The answer is: Expected Output: Eintracht Frankfurt

Table 9: Our prompt for prompting the pretrained LM using the sensitive w/o time strategy for time-sensitive QA tasks.

Prompt: Answer the following question: Which Hindi film has the highest domestic net collection currently? As of year 2022, the answer is: Brahmāstra: Part One – Shiva Answer the following question: Where is the NHL Winter Classic taking place? As of year 2022, the answer is: Target Field Answer the following question: Who are the current drivers for the Mercedes-Benz Formula One team? As of year 2022, the answer is: Lewis Hamilton George Russell Answer the following question: Who received the Player of the Game award for offense in the most recent Rose Bowl Game? As of year 2022, the answer is: Jaxon Smith-Njigba Answer the following question: Where was the final of the last FIFA Club World Cup held? As of year 2022, the answer is: Prince Moulay Abdellah Stadium, Rabat Answer the following question: Which team won the UEFA Europa League? As of year 2022, the answer is: Expected Output: Eintracht Frankfurt
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Table 10: Our prompt for prompting the pretrained LM using the sensitive w/ time strategy for time-sensitive QA tasks.